

Elementary Problems

Congruence

Solutions

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Basic Work

1. **Solution 1:**

Since $n^3 - n = (n - 1)n(n + 1)$ is a multiple of three consecutive integers, we have $6 \mid n^3 - n$. In other words, $n^3 \equiv n \pmod{6}$.

Solution 2:

Since $n \equiv r \pmod{6}$ implies $n^3 - n \equiv r^3 - r \pmod{6}$, it suffices to check the cases $n = 0, 1, 2, 3, 4, 5$. We find that $n^3 - n \equiv 0 \pmod{6}$ for all these cases.

2. We have $3n \equiv 2 \equiv 9 \pmod{7}$. As $(3, 7) = 1$, we can divide both sides by 3. This gives the solution $n \equiv 3 \pmod{7}$.
3. From $4n \equiv 1 \pmod{6}$, we get $4n \equiv 1 \pmod{2}$. However, this yields the contradiction $0 \equiv 1 \pmod{2}$. Thus, there is no solution.
4. We have $15n \equiv 5 \equiv 15 \pmod{10}$. As $(15, 10) = 5$, this implies

$$n \equiv 1 \pmod{\frac{10}{5}},$$

i.e. $n \equiv 1 \pmod{2}$. The steps are reversible and hence the solution is any odd integer n .

5. If $p = 3$, then $p + 2 = 5$ and $p + 4 = 7$ are primes.

If $p = 3k + 1$, then $p + 2 = 3k + 3$ is divisible by 3. As $p + 2 \geq 2 + 2 = 4$, it cannot be a prime.

If $p = 3k + 2$, then $p + 4 = 3k + 6$ is divisible by 3. As $p + 4 > 4$, it cannot be a prime.

6. Since $(3n - 4) + (4n - 5) + (5n - 3) = 12n - 12$ is even, at least one of the prime numbers is even. Thus, one of them equals to 2.

If $3n - 4 = 2$, then $n = 2$. We check that $3n - 4 = 2$, $4n - 5 = 3$ and $5n - 3 = 7$ are primes.

If $4n - 5 = 2$, then there is no solution.

If $5n - 3 = 2$, then $n = 1$. However, $3n - 4 = -1$ is not a prime.

Thus, the only solution is $n = 2$.

7. If $n = 2k$, then $2^{2k} - 1 = 4^k - 1 \equiv 1 - 1 = 0 \pmod{3}$. The only possibility is $2^n - 1 = 3$, i.e. $n = 2$. In that case, both $2^n - 1 = 3$ and $2^n + 1 = 5$ are prime numbers.

If $n = 2k - 1$, then $2^{2k-1} + 1 = 2 \cdot 4^{k-1} + 1 \equiv 2 + 1 = 0 \pmod{3}$. The only possibility is $2^n + 1 = 3$, i.e. $n = 1$. However, $2^n - 1 = 1$ is not a prime.

Therefore, the only solution is $n = 2$.

8. Let the decimal representation of n be $\overline{a_d a_{d-1} \cdots a_0}$. Then

$$n = a_d \times 10^d + a_{d-1} \times 10^{d-1} + \cdots + a_0.$$

Taking modulo 9, as $10 \equiv 1 \pmod{9}$, we have

$$n \equiv a_d + a_{d-1} + \cdots + a_0 \pmod{9}.$$

This is the result we need.

9. Let the sum of the digits of B be C . Note that

$$A \equiv 4444^{4444} \equiv 7^{4444} \equiv 7 \pmod{9}$$

since the remainder of an integer n when divided by 9 is the same as that when the sum of digits of n is divided by 9. Also, we have used the fact that $7^3 \equiv 1 \pmod{9}$ and $4444 \equiv 1 \pmod{3}$. Similarly, we have $C \equiv B \equiv A \equiv 7 \pmod{9}$.

Next, $4444^{4444} < 10000^{4444} = 10^{17776}$ consists of at most 17776 digits. Thus, we have $A \leq 9 \cdot 17776 < 180000$. This implies $B \leq 9 \cdot 5 = 45$, and hence $C \leq 3 + 9 = 12$. The only possibility for $C \equiv 7 \pmod{9}$ to hold is $C = 7$.

10. Firstly, since $2^{1990} < 10^{1990}$, we have

$$f(2^{1990}) \leq (9 \times 1990)^2 < 10^{10}.$$

Secondly, this implies

$$f_2(2^{1990}) \leq (9 \times 10)^2 = 8100.$$

Similarly, we find that

$$f_3(2^{1990}) \leq (7 + 9 + 9 + 9)^2 = 1156$$

and

$$f_4(2^{1990}) \leq (9 \times 3)^2 = 729.$$

It follows that the sum of digits of $f_k(2^{1990})$ cannot exceed 27 for $k \geq 4$.

Next, we have $2^{1990} \equiv 2^4 \equiv 7 \pmod{9}$. We easily find that $f_k(2^{1990}) \equiv 4 \pmod{9}$ for odd k and $f_k(2^{1990}) \equiv 7 \pmod{9}$ for even k . Thus, the sum of digits of $f_{1987}(2^{1990})$ can only be 4, 13, 22. This yields $f_{1988}(2^{1990}) = 16, 169, 484$, $f_{1989}(2^{1990}) = 49, 256$, $f_{1990}(2^{1990}) = 169$ and hence $f_{1991}(2^{1990}) = 256$.

11. No. By testing $n \equiv 0, 1, 2 \pmod{3}$, we have $n^2 + n \equiv 0, 2 \pmod{3}$. This implies

$$3n^2 + 3n + 7 = 3(n^2 + n) + 7 \equiv 4, 7 \pmod{9}.$$

However, for any integer m , we have $m^3 \equiv -1, 0, 1 \pmod{9}$. This shows $3n^2 + 3n + 7$ cannot be a cube.

12. Suppose such a solution exists. This means $x^3 + y^3 + z^3 = 1969^2$ for some integers x, y, z . Note that $m^3 \equiv -1, 0, 1 \pmod{9}$ for any integer m . From this, we find that $x^3 + y^3 + z^3 \equiv -3, -2, -1, 0, 1, 2, 3 \pmod{9}$. However, we have $1969^2 \equiv 4 \pmod{9}$. This is a contradiction. Therefore, there is no integer solution.
13. If n is even, then $n^4 \equiv 0 \pmod{16}$. If n is odd, then since $n^2 \equiv 1 \pmod{8}$, we have $n^2 = 8k + 1$ for some integer k , and so $n^4 = 64k^2 + 16k + 1 \equiv 1 \pmod{16}$. Hence,

$$n_1^4 + n_2^4 + \cdots + n_{14}^4 \equiv 0, 1, 2, \dots, 14 \pmod{16}.$$

As $1599 \equiv 15 \pmod{16}$, there is no integer solution.

14. Consider modulo 13. One checks that

$$m^3 \equiv 0, \pm 1, \pm 5 \pmod{13}$$

and

$$n^4 \equiv 0, 1, 3, 9 \pmod{13}.$$

These show

$$m^3 + n^4 \equiv 0, 1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12 \not\equiv 7 \pmod{13}.$$

Hence, there is no solution.

15. Consider modulo 11. One checks that $m^2 \equiv 0, 1, 3, 4, 5, 9 \pmod{11}$. Therefore,

$$m^2 + 4 \equiv 2, 4, 5, 7, 8, 9 \pmod{11}.$$

On the other hand, we find that

$$n^5 \equiv -1, 0, 1 \pmod{11}.$$

These show $m^2 + 4 \not\equiv n^5 \pmod{11}$. Thus, there is no integer solution.

16. Suppose on the contrary that $(p, d) = 1$. Then the linear congruence equation

$$a + kd \equiv 0 \pmod{p}$$

has a unique solution in k , and we can find such a solution in the range $1 \leq k \leq p$. This contradicts the fact that $a + kd$ is not divisible by p . Therefore, we must have $p \mid d$.

If $p \mid a$, then $p \mid a + d$, which is a contradiction. Thus, we have $p \nmid a$.

17. Let the arithmetic progression be $a + kd$ where $k = 0, 1, 2, \dots$. It follows that the arithmetic progression contains all sufficiently large integer $n \equiv a \pmod{d}$. It is given that there exists m such that $m^2 \equiv a \pmod{d}$. As $(m + \ell d)^2 \equiv m^2 \equiv a \pmod{d}$ for each $\ell = 0, 1, 2, \dots$, all the square numbers $(m + \ell d)^2$ belong to the arithmetic progression.

18. We consider every pair of consecutive terms modulo p . Since there are only p^2 possibilities for the pair of remainders, there must be two pairs which are the same. Whenever we have $(F_n, F_{n+1}) \equiv (F_m, F_{m+1}) \pmod{p}$ for some $m \neq n$, we have

$$F_{n+2} = F_{n+1} + F_n \equiv F_{m+1} + F_m = F_{m+2} \pmod{p}$$

and

$$F_{n-1} = F_{n+1} - F_n \equiv F_{m+1} - F_m = F_{m-1} \pmod{p}.$$

Inductively, this shows the remainders of the terms modulo p is periodic. In particular, there are infinitely many $n \in \mathbb{N}$ such that $F_{n+1} \equiv F_n \equiv 1 \pmod{p}$. Thus,

$$F_{n-1} = F_{n+1} - F_n \equiv 1 - 1 = 0 \pmod{p}$$

for infinitely many n .

19. Consider modulo 8. The sequence (X_n) has the pattern

$$X_n \equiv 1, 1, 3, 5, 3, 5, \dots \pmod{8}$$

while the sequence (Y_n) has the pattern

$$Y_n \equiv 1, 7, 1, 7, 1, 7, \dots \pmod{8}.$$

Therefore, the only possible terms which may occur in both sequences are X_1 and X_2 , which are equal to 1.

20. If a_1 is even, then $a_2 \equiv 3 \pmod{4}$ and $a_3 \equiv 2 \pmod{4}$. Inductively, we can show that $a_k \equiv 3 \pmod{4}$ if k is even and $a_k \equiv 2 \pmod{4}$ if $k \geq 3$ is odd.

If $a_1 \equiv 3 \pmod{4}$, then $a_k \equiv 3 \pmod{4}$ if k is odd and $a_k \equiv 2 \pmod{4}$ if k is even as shown above.

If $a_1 \equiv 1 \pmod{4}$, then $a_2 \equiv 0 \pmod{4}$. The above shows that $a_k \equiv 3 \pmod{4}$ if $k \geq 3$ is odd and $a_k \equiv 2 \pmod{4}$ if $k \geq 4$ is even.

Since a square number is congruent to 0, 1 modulo 4, the only possible perfect squares are a_1 and a_2 . Suppose on the contrary that both of them are squares. Let $a_1 = b^2$ and $a_2 = c^2$. Then $c^2 = b^6 + 11$. This yields

$$11 = c^2 - b^6 = (c - b^3)(c + b^3).$$

The only possibility is $c - b^3 = 1$ and $c + b^3 = 11$. However, this gives no integer solution. Therefore, there can be at most one perfect square.

21. Let $P(1) = p$ be a prime. For each $k \in \mathbb{N}$, since

$$P(kp + 1) \equiv P(1) \equiv 0 \pmod{p}$$

and $P(kp + 1)$ is a prime, we must have $P(kp + 1) = p$ for all k . As this holds for infinitely many k 's, $P(n)$ must be a constant polynomial.

22. Firstly, if each of $\{a_1, a_2, \dots, a_n\}$, $\{b_1, b_2, \dots, b_n\}$ and $\{a_1 + b_1, a_2 + b_2, \dots, a_n + b_n\}$ is a complete residue system, then the sum of elements in each set is congruent to

$$1 + 2 + \dots + n = \frac{n(n+1)}{2} \pmod{n}.$$

Therefore, we have $\frac{n(n+1)}{2} + \frac{n(n+1)}{2} \equiv \frac{n(n+1)}{2} \pmod{n}$, which gives $n \mid \frac{n(n+1)}{2}$. This holds if and only if n is odd.

If n is odd, then such sets exist. For example, we can take $a_j = b_j = j$ for $j = 1, 2, \dots, n$. In that case, we have $a_j + b_j = 2j$. The set $\{2, 4, \dots, 2n\}$ is a complete residue system modulo n . Indeed, the terms $2j \pmod{n}$ covers all residue classes since $(2, n) = 1$.

Therefore, n can be any odd positive integer.

23. For $i > k$, the numbers i and $i - k$ have the same colour. This shows the colour of a number depends only on its remainder when divided by k . Therefore, it suffices to show $1, 2, \dots, k$ have the same colour.

Since $(k, n) = 1$, the term $-mn$ covers all residue classes modulo k when m runs from 1 to k . Let $r_1, r_2, \dots, r_k$ be the permutation of $1, 2, \dots, k$ such that $r_m \equiv -mn \pmod{k}$ for $1 \leq m \leq k$.

WLOG assume r_1 is blue. We prove by induction that r_m is blue for $1 \leq m \leq k$. Assume r_m is blue for some $1 \leq m \leq k - 1$. Then $n - r_m$ is blue. Let s be the unique number among $1, 2, \dots, k$ such that $n - r_m \equiv s \pmod{k}$. Then s is blue by subtracting sufficiently many k 's from $n - r_m$. Note that $s \neq k$ unless $m = k - 1$, since otherwise we have

$$0 \equiv n - r_m \equiv n + mn = n(1 + m) \pmod{k},$$

which cannot be true as $(n, k) = 1$ and $0 < 1 + m < k$. Thus, we have $1 \leq s \leq k - 1$. This implies $k - s$ is blue. Note that

$$k - s \equiv 0 - (n - r_m) = r_m - n \equiv r_{m+1} \pmod{k}.$$

As $1 \leq s \leq k - 1$, this gives $r_{m+1} = k - s$ is blue. By repeating the inductive step, we can show that $r_1, r_2, \dots, r_{k-1}$ are blue. When $m = k - 1$, the above s becomes k . This means $r_k = k$ is also blue.

Therefore, all numbers in M have the same colour.

24. (a) False. For example, if $a = b = 2$ and $n = 4$, then $ab \equiv 0 \pmod{n}$ but neither a nor b is congruent to 0 modulo n .
- (b) False. For example, if $a = 1$, $b = 3$ and $n = 8$, then $a^2 = 1 \equiv b^2 \pmod{8}$, but $a \not\equiv \pm b \pmod{8}$.
- (c) False. For example, if $a = b = c = 1$, $d = 3$ and $m = n = 2$, then a, b, c, d have the same parity, but $ac = 1 \not\equiv 3 = bd \pmod{mn}$.

- (d) True. Firstly, we have $(m, m+n) = (m, n) = 1$. Similarly, we have $(n, m+n) = 1$. Thus, mn is relatively prime to $m+n$. It follows that the linear equation has a unique solution modulo $m+n$.
- (e) False. For example, if $P(x) = x^2 - 1$ and $n = 8$, then $d = 2$ but $m^2 - 1 \equiv 0 \pmod{8}$ has 4 solutions $m \equiv 1, 3, 5, 7 \pmod{8}$.

Chinese Remainder Theorem

25. From $n \equiv 5 \pmod{7}$, we have $n = 5, 12, 19, \dots$. Note that $n = 5$ also satisfies the congruence $n \equiv 2 \pmod{3}$. By the Chinese remainder theorem, there is a unique solution modulo $3 \times 7 = 21$. Thus, the solution is $n \equiv 5 \pmod{21}$.
26. From $3n \equiv 1 \equiv 6 \pmod{5}$, we have $n \equiv 2 \pmod{5}$.
 From $2n \equiv 4 \pmod{8}$, we have $n \equiv 2 \pmod{4}$ (but not modulo 8). Clearly, $n = 2$ is a solution. By the Chinese remainder theorem, there is a unique solution modulo $5 \times 4 = 20$. Thus, the solution is $n \equiv 2 \pmod{20}$.
27. From $-n \equiv 2n \equiv 1 \pmod{3}$, we have $n \equiv -1 \pmod{3}$.
 From $-n \equiv 3n \equiv 1 \pmod{4}$, we have $n \equiv -1 \pmod{4}$.
 From $2n \equiv 3 \equiv 8 \pmod{5}$, we have $n \equiv 4 \equiv -1 \pmod{5}$. Clearly, $n = -1$ is a solution. By the Chinese remainder theorem, there is a unique solution modulo $3 \times 4 \times 5 = 60$. Thus, the solution is $n \equiv -1 \pmod{60}$.
28. We have $n^2 \equiv 144 = 12^2 \pmod{1000}$. This is the same as

$$\begin{aligned} n^2 &\equiv 12^2 \pmod{8}, \\ n^2 &\equiv 12^2 \pmod{125}. \end{aligned}$$

Firstly, $n^2 \equiv 12^2 \equiv 0 \pmod{8}$ implies $8 \mid n^2$. Hence, we have $4 \mid n$.

Secondly, $n^2 \equiv 12^2 \pmod{125}$ implies $125 \mid (n-12)(n+12)$. As

$$(n-12, n+12) = (n-12, 24),$$

exactly one of $n-12$ and $n+12$ can be a multiple of 5. Thus, $n \equiv \pm 12 \pmod{125}$. Solving

$$\begin{aligned} n &\equiv 0 \pmod{4}, \\ n &\equiv \pm 12 \pmod{125}, \end{aligned}$$

we get $n \equiv \pm 12 \pmod{500}$ by the Chinese remainder theorem. Therefore, n can be any positive integer of the form $500k \pm 12$.

29. Let $p_1, p_2, \dots, p_n, q_1, q_2, \dots, q_n$ be distinct primes. By the Chinese remainder theorem, there exists a positive integer m such that

$$m \equiv -k \pmod{p_k q_k}$$

for $k = 1, 2, \dots, n$. For this m , each $m + k$ has two distinct prime divisors so that it cannot be a power of prime. Thus, none of $m + 1, m + 2, \dots, m + n$ is a power of prime.

30. Firstly, if $k \equiv 2 \pmod{3}$, then $p^2 + k \equiv 1 + k \equiv 0 \pmod{3}$ for any $p \neq 3$. Secondly, if $k \equiv 1 \pmod{5}$, then $3^2 + k \equiv 0 \pmod{5}$. Thus, it suffices to choose those positive integers k satisfying

$$\begin{aligned} k &\equiv 0 \pmod{2}, \\ k &\equiv 2 \pmod{3}, \\ k &\equiv 1 \pmod{5}. \end{aligned}$$

There are infinitely many such k 's by the Chinese remainder theorem. For each such k , $p^2 + k$ is not a prime from above.

31. Let $a_i = 1 + (i - 1)k$ for $i = 1, 2, \dots, m$ and $b_j = 1 + (j - 1)m$ for $j = 1, 2, \dots, k$. For any $n \pmod{mk}$, since $(m, k) = 1$, there exist x and y such that

$$n \equiv 1 + (x - 1)k \pmod{m} \quad \text{and} \quad n \equiv 1 + (y - 1)m \pmod{k}.$$

Then we have

$$a_x b_y = (1 + (x - 1)k)(1 + (y - 1)m) \equiv 1 + (x - 1)k \equiv n \pmod{m}$$

and similarly $a_x b_y \equiv n \pmod{k}$. These imply $a_x b_y \equiv n \pmod{mk}$ by the Chinese remainder theorem as $(m, k) = 1$. Thus, the product $a_i b_j$ covers all residue classes modulo mk .

32. By the Chinese remainder theorem, the divisibility is the same as

$$\begin{aligned} 9^{p^2} - 29^p + m &\equiv 0 \pmod{3}, \\ 9^{p^2} - 29^p + m &\equiv 0 \pmod{5}, \\ 9^{p^2} - 29^p + m &\equiv 0 \pmod{7}. \end{aligned}$$

The first congruence is the same as $0 - (-1)^p + m \equiv 0 \pmod{3}$. Since p is an odd prime, this becomes $m \equiv -1 \pmod{3}$.

The second congruence is the same as $(-1)^{p^2} - (-1)^p + m \equiv 0 \pmod{5}$. As p is odd, this becomes $m \equiv 0 \pmod{5}$.

For the third congruence, we note from $p^2 \equiv 1 \pmod{3}$ and $9^3 \equiv 2^3 \equiv 1 \pmod{7}$ that

$$9^{p^2} - 29^p + m \equiv 9 - 1 + m \equiv 1 + m \pmod{7}.$$

Hence, we need $m \equiv -1 \pmod{7}$.

By the Chinese remainder theorem, there exists a unique solution of m modulo 105. Indeed, from $m \equiv -1 \pmod{3}$ and $m \equiv -1 \pmod{7}$, we have $m \equiv -1 \pmod{21}$. It is not hard to see that $m \equiv 20 \pmod{105}$ satisfies all congruences. Hence, the smallest m is 20.

33. Since there are 100 consecutive integers, they must cover all residue classes modulo 100 once. Thus,

$$a_1^8 + a_2^8 + \cdots + a_{100}^8 \equiv 1^8 + 2^8 + \cdots + 100^8 \pmod{100}.$$

We consider modulo 4 and modulo 25 separately. Firstly,

$$1^8 + 2^8 + \cdots + 100^8 \equiv 25(1^8 + 2^8 + 3^8 + 4^8) \equiv 25(1 + 0 + 1 + 0) \equiv 2 \pmod{4}.$$

Secondly,

$$1^8 + 2^8 + \cdots + 100^8 \equiv 4(1^8 + 2^8 + \cdots + 25^8) \pmod{25}.$$

For each $k = 1, 2, 3, 4$, using binomial expansion, we have

$$\begin{aligned} & k^8 + (k+5)^8 + (k+10)^8 + (k+15)^8 + (k+20)^8 \\ & \equiv k^8 + (k^8 + 8k^7 \cdot 5) + (k^8 + 8k^7 \cdot 10) + (k^8 + 8k^7 \cdot 15) + (k^8 + 8k^7 \cdot 20) \\ & \equiv 5k^8 \pmod{25}. \end{aligned}$$

Summing over these k 's, we find that

$$1^8 + 2^8 + \cdots + 100^8 \equiv 4 \times 5(1^8 + 2^8 + 3^8 + 4^8) \equiv 20(1 + 6 + 11 + 11) \equiv 5 \pmod{25}.$$

Thus, the sum S satisfies $S \equiv 2 \pmod{4}$ and $S \equiv 5 \pmod{25}$. The Chinese remainder theorem asserts that there is a unique solution modulo 100. Indeed, from the congruence modulo 25, it suffices to test $S = 5, 30, 55, 80$. We find that $S \equiv 30 \pmod{100}$. This shows the last two digits of S are 30.

Euler-Fermat Theorem

34. By Fermat little theorem, we have $2^{16} \equiv 1 \pmod{17}$ since $(2, 17) = 1$. Thus,

$$2^{1000} = (2^{62})^{16} 2^8 \equiv 2^8 = 256 \equiv 1 \pmod{17}.$$

So the remainder is 1.

35. By Euler-Fermat theorem, we have $3^{40} \equiv 1 \pmod{100}$ as $(3, 100) = 1$ and $\varphi(100) = 40$. Thus,

$$3^{1234} = (3^{40})^{31} 3^{-6} \equiv 3^{-6} \pmod{100}.$$

Here, 3^{-1} means the inverse of 3 modulo 100. Solving $3x \equiv 1 \equiv 201 \pmod{100}$, we easily get $x \equiv 67 \pmod{100}$. Thus,

$$3^{-6} \equiv 67^6 = 4489^3 \equiv (-11)^3 = -1331 \equiv 69.$$

This means $3^{1234} \equiv 69 \pmod{100}$.

36. Note that $221 = 13 \times 17$. It suffices to consider $5^{50} \pmod{13}$ and $5^{50} \pmod{17}$.

By Fermat little theorem, we have $5^{12} \equiv 1 \pmod{13}$. Thus,

$$5^{50} = (5^{12})^4 5^2 \equiv 5^2 = 25 \pmod{13}.$$

Similarly, we have

$$5^{50} = (5^{16})^3 5^2 \equiv 5^2 = 25 \pmod{17}.$$

By the Chinese remainder theorem, these yield $5^{50} \equiv 25 \pmod{221}$. So the remainder is 25.

37. Firstly, we choose $a_1 = 2^3 - 3$. Suppose $a_1, a_2, \dots, a_n > 3$ have been chosen, each of which has the form $2^k - 3$ and every pair of which are relatively prime. Then we choose

$$a_{n+1} = 2^{\varphi(a_1)\varphi(a_2)\cdots\varphi(a_n)} - 3.$$

Since each $a_j > 3$, we have $a_{n+1} > 3$. Also, since a_j is odd, we have $2^{\varphi(a_j)} \equiv 1 \pmod{a_j}$ by Euler-Fermat theorem. Thus,

$$a_{n+1} \equiv 1 - 3 = -2 \pmod{a_j}.$$

This shows $(a_{n+1}, a_j) = (-2, a_j) = 1$ for each $j = 1, 2, \dots, n$. In particular, a_{n+1} is distinct from each a_j . This recursively defines an infinite sequence of numbers of the form $2^k - 3$ such that every two of them are relatively prime.

38. We choose $1 \leq r \leq 9$ such that $p \equiv r \pmod{10}$. By Fermat little theorem, we have

$$2^p + 3^p \equiv 2^r + 3^r \pmod{11}$$

since $(2, 11) = (3, 11) = 1$. We check each case as follows:

$$\begin{aligned} 2^1 + 3^1 &\equiv 5 \pmod{11}, \\ 2^2 + 3^2 &\equiv 4 + 9 \equiv 2 \pmod{11}, \\ 2^3 + 3^3 &\equiv 8 + 5 \equiv 2 \pmod{11}, \\ 2^4 + 3^4 &\equiv 5 + 4 \equiv 9 \pmod{11}, \end{aligned}$$

$$\begin{aligned}
2^5 + 3^5 &\equiv 10 + 1 \equiv 0 \pmod{11}, \\
2^6 + 3^6 &\equiv 9 + 3 \equiv 1 \pmod{11}, \\
2^7 + 3^7 &\equiv 7 + 9 \equiv 5 \pmod{11}, \\
2^8 + 3^8 &\equiv 3 + 5 \equiv 8 \pmod{11}, \\
2^9 + 3^9 &\equiv 6 + 4 \equiv 10 \pmod{11}.
\end{aligned}$$

These show the only case that $11 \mid 2^r + 3^r$ is $r = 5$. Therefore, $p \equiv 5 \pmod{10}$. This implies $5 \mid p$. As p is a prime, we must have $p = 5$. This is a solution from above.

39. Suppose on the contrary that $p \mid 2^{p-1} + 3^{p-1}$. Clearly, $p \neq 2, 3$. Thus, $(2, p) = (3, p) = 1$. By Fermat little theorem, we have

$$2^{p-1} + 3^{p-1} \equiv 1 + 1 = 2 \pmod{p}.$$

Since $p > 2$, this is nonzero modulo p . Thus, $2^{p-1} + 3^{p-1}$ is not divisible by p .

40. If $p = q$, then we need $p^2 \mid 2p^p + 7$. In particular, $p \mid 2p^p + 7$, and hence $p \mid 7$. The only possibility is $p = 7$. This is rejected since $7^2 \nmid 2 \cdot 7^7 + 7$.

WLOG assume $p > q$. From $p \mid p^q + q^p + 7$, we have $p \mid q^p + 7$. Using Fermat little theorem, this implies $p \mid q + 7$. If $p = q + 7$, then $q = 2$ since p is odd and q is a prime. However, $p = 9$ is not a prime. Hence, we have $p \leq \frac{q+7}{2}$. As $p \geq q + 1$, we get $q + 7 \geq 2p \geq 2(q + 1)$. This implies $q \leq 5$.

If $q = 5$, then $p \mid 12$. There is no prime $p > 5$ which divides 12. If $q = 3$, then $p \mid 10$. The only possibility is $p = 5$. One checks that $3^5 + 5^3 + 7 = 375$ is divisible by $3 \cdot 5 = 15$. If $q = 2$, then $p \mid 9$. The only possibility is $p = 3$. One checks that $2^3 + 3^2 + 7 = 24$ is divisible by $2 \cdot 3 = 6$.

Thus, the only solutions are $(p, q) = (2, 3), (3, 2), (3, 5), (5, 3)$.

41. Taking modulo 3, by Fermat little theorem, we have

$$m - n \equiv m^3 - n^3 \equiv 0 \pmod{3}.$$

Let $m = n + 3k$. Then

$$m^3 - n^3 = (n + 3k)^3 - n^3 \equiv n^3 + \binom{3}{1}n^2(3k) - n^3 \equiv 0 \pmod{9}$$

by the binomial theorem. This contradicts the fact that $9 \nmid 2013$.

Hence, there is no integer solution.

42. (a) Let $n = p$ be a prime. If $p \mid a^p - 1$, then we have

$$a - 1 \equiv a^p - 1 \equiv 0 \pmod{p}$$

by Fermat little theorem. Let $a = pm + 1$ for some $m \in \mathbb{Z}$. By the binomial theorem, we have

$$a^p - 1 = (pm + 1)^p - 1 \equiv 1 + \binom{p}{1}(pm) - 1 \equiv 0 \pmod{p^2}$$

as $(pm)^j \equiv 0 \pmod{p^2}$ for $j \geq 2$. This implies $p^2 \mid a^p - 1$. So p has property P .

- (b) We claim that any $n = pq$ where p, q are distinct primes possesses property P . If $pq \mid a^{pq} - 1$, then we have $p \mid a^{pq} - 1$. By Fermat little theorem, we get

$$a^q - 1 \equiv a^{pq} - 1 \equiv 0 \pmod{p}.$$

Let $a^q = pm + 1$ for some $m \in \mathbb{Z}$. By the binomial theorem, we have

$$a^{pq} - 1 = (pm + 1)^p - 1 \equiv 1 + \binom{p}{1}(pm) - 1 \equiv 0 \pmod{p^2}.$$

This implies $p^2 \mid a^{pq} - 1$. Similarly, we have $q^2 \mid a^{pq} - 1$. As p and q are distinct primes, these yield $p^2 q^2 \mid a^{pq} - 1$. So pq has property P .

43. Consider any $n \in \mathbb{N}$ such that

$$\begin{aligned} n &\equiv 1 \pmod{p-1}, \\ n &\equiv 2 \pmod{p}. \end{aligned}$$

There are infinitely many such n 's by the Chinese remainder theorem as $(p-1, p) = 1$. For any such n , we have

$$2^n - n \equiv 2^1 - n \equiv 2 - 2 = 0 \pmod{p}$$

by Fermat little theorem. This shows $p \mid 2^n - n$.

44. As m and n are relatively prime, by Euler-Fermat theorem, we have $m^{\varphi(n)} \equiv 1 \pmod{n}$ and $n^{\varphi(m)} \equiv 1 \pmod{m}$. We take $k = \varphi(m)\varphi(n)$. Then

$$m^k + n^k - 1 \equiv 0 + (n^{\varphi(m)})^{\varphi(n)} - 1 \equiv 1^{\varphi(n)} - 1 = 0 \pmod{m}.$$

Similarly $m^k + n^k - 1 \equiv 0 \pmod{n}$. As $(m, n) = 1$, we have $mn \mid m^k + n^k - 1$.

45. Since

$$a_1 \cdot 1 + a_2 \cdot 2 + \cdots + a_m \cdot m > 1,$$

we can pick a prime divisor p of this number. For any $n \equiv 1 \pmod{p-1}$, we have

$$a_1 \cdot 1^n + a_2 \cdot 2^n + \cdots + a_m \cdot m^n \equiv a_1 \cdot 1 + a_2 \cdot 2 + \cdots + a_m \cdot m \pmod{p}$$

by Fermat little theorem. This is divisible by p from the construction. Clearly, this is larger than p for $n > 1$. Hence, there are infinitely many n 's satisfying the condition.

46. By the Chinese remainder theorem, there exists an integer n such that

$$\begin{cases} n \equiv 1 \pmod{p}, \\ n \equiv 0 \pmod{p-1}. \end{cases}$$

For this choice of n , since $p > k+1$ and $n+j \equiv 1+j \pmod{p}$ for each $j = 0, 1, \dots, k$, we have $(n+j, p) = 1$. Therefore, from $n \equiv 0 \pmod{p-1}$ and by Fermat little theorem, we get $(n+j)^n \equiv 1 \pmod{p}$. This shows

$$(n+j)^n + j \equiv 1 + j \pmod{p},$$

which cannot be congruent to 0 modulo p as $1 \leq 1+j \leq k+1 < p$. Thus, none of

$$n^n, (n+1)^n + 1, (n+2)^n + 2, \dots, (n+k)^n + k$$

is divisible by p .

47. The case $q = 1$ is trivial.

We first assume q is a prime. If q divides one of n and $n+1$, then it divides both of them, which is impossible. Thus, we may assume $(q, n) = (q, n+1) = 1$.

If $(p, q-1) = 1$, then there exists a positive integer m such that $pm \equiv 1 \pmod{q-1}$. By Fermat little theorem, we have

$$n+1 \equiv (n+1)^{pm} \equiv n^{pm} \equiv n \pmod{q}.$$

This is a contradiction. Therefore, we must have $p \mid q-1$. This shows the prime q is of the form $kp+1$.

In general, for any divisor q of $(n+1)^p - n^p$, the above shows that any prime divisor of q is of the form $kp+1$. Hence, q is also of the form $kp+1$. In other words, $p \mid q-1$.

48. By Dirichlet's theorem, there exist infinitely many primes p of the form $6k+5$. We can pick a sufficiently large $p > n$ so that $p \nmid n$. It remains to show $n \equiv m^3 \pmod{p}$ for some integer m .

Indeed, by Fermat little theorem, we have $n^{6k+4} = n^{p-1} \equiv 1 \pmod{p}$. Hence, we have

$$n \equiv n^{2(6k+4)+1} = n^{12k+9} = (n^{4k+3})^3 \pmod{p}.$$

Hence, we can take $m = n^{4k+3}$. This completes the proof.